

Notes: Tetlock, Saar-Tsechansky & Macskassy (JF, 2008)

More Than Words: Quantifying Language to Measure Firms' Fundamentals

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1 Big picture

- **Question.** Can a simple dictionary-based measure of firm-specific news language predict firms' future earnings and stock returns?
- **Core object.** The paper measures the fraction of negative words in WSJ and DJNS stories about S&P 500 firms from 1980 to 2004.
- **Main earnings result.** Negative words forecast lower standardized unexpected earnings and lower analyst forecast errors.
- **Main return result.** Prices react quickly on the news day but exhibit a small one-day delayed reaction, especially for DJNS stories.
- **Mechanism interpretation.** Predictability is strongest when stories mention the word stem “earn,” suggesting the language variable captures hard-to-quantify fundamentals.
- **Takeaway.** Qualitative media content can be transformed into an objective quantitative variable that helps measure firm fundamentals and price discovery.

2 Incremental contribution of the paper

- It moves textual analysis from aggregate sentiment to firm-level fundamentals.
- It connects words to cash-flow outcomes by showing that negative language predicts earnings, not just returns.
- It analyzes everyday news across many event types, reducing event-selection bias.
- It introduces a transparent, replicable dictionary-based benchmark for later textual-finance research.
- It distinguishes generic negative tone from fundamentals-related tone using stories that mention the word stem “earn.”
- It shows near-efficiency rather than gross inefficiency: prices mostly react immediately, with only a small delayed component.

3 Data and measurement

3.1 News stories

The sample consists of WSJ and DJNS stories about S&P 500 firms from 1980 to 2004. Stories are combined into firm-day composite stories before counting words.

Interpretation: This captures a broad set of firm-specific qualitative information rather than one selected corporate event.

3.2 Negative-word measure

$$Neg = \frac{\# \text{ negative words}}{\# \text{ total words}}, \quad neg = \frac{Neg - \mu_{Neg}}{\sigma_{Neg}},$$

where μ_{Neg} and σ_{Neg} are computed from the previous calendar year's distribution.

Interpretation: Standardization controls for changing media style, coverage, and vocabulary over time.

3.3 Earnings and returns

$$UE_t = E_t - E_{t-4}, \quad SUE_t = \frac{UE_t - \mu_{UE}}{\sigma_{UE}}.$$

The paper also uses standardized analyst forecast errors (SAFE), close-to-close returns, and Fama–French abnormal returns.

Interpretation: SUE and SAFE test whether language predicts fundamentals; returns test whether prices fully incorporate the linguistic signal.

4 Models and empirical specifications

4.1 Earnings predictability

$$Earnings_{it} = \alpha + \beta neg_{i,-30,-3} + \gamma' Controls_{it} + \varepsilon_{it}.$$

Interpretation: A negative β means that more negative language before an earnings announcement predicts lower realized earnings.

4.2 Return predictability

$$Return_{i,t+1} = \alpha + \beta neg_{it} + \gamma' Controls_{it} + \varepsilon_{it}.$$

Interpretation: A negative β implies delayed price adjustment to the information embedded in news language.

4.3 Fundamentals interaction

$$Y = \alpha + \beta_1 neg + \beta_2 Fund + \beta_3 (neg \times Fund) + \gamma' Controls + \varepsilon.$$

Interpretation: β_3 tests whether negative words are especially informative when the story is about earnings or fundamentals.

5 Identification and empirical design

This is a predictive and market-efficiency design, not a randomized causal design. The paper uses timing, controls, robustness tests, and heterogeneity to interpret linguistic predictability.

- **Timing discipline.** Earnings tests use stories from days -30 to -3 before earnings announcements.
- **Cash-flow validation.** Negative words predict SUE and SAFE, supporting a fundamentals interpretation.
- **Controls.** Regressions include analyst variables, lagged earnings, firm characteristics, and past returns.
- **Before-forecast test.** Negative words remain predictive when measured before the latest analyst forecast.
- **Return underreaction test.** Next-day return predictability tests whether prices fully incorporate the information.

- **Source timing.** Stronger DJNS predictability is consistent with short-run processing frictions in a continuous news source.
- **Fundamentals heterogeneity.** Earnings-related stories predict more strongly, supporting the cash-flow channel.
- **Trading cost discipline.** Profitability is sensitive to transaction costs, so the evidence implies small underreaction rather than easy arbitrage.

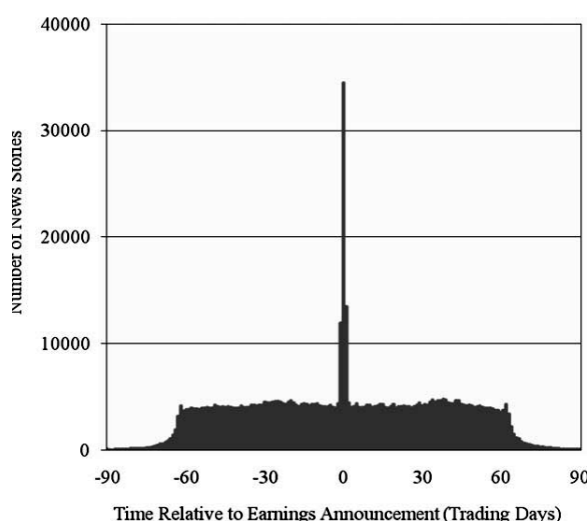
Interpretation: The design is credible because the same language measure predicts fundamentals, predicts short-run returns, and is strongest in stories where fundamentals are most likely to be discussed.

6 Tables and figures

6.1 Figure 1: Media Coverage around Earnings Announcements; Table I: Predicting Earnings Using Negative Words

Read: Figure 1 shows that news stories cluster around earnings announcements. Table I shows that negative words predict lower SUE and SAFE across sources and robustness windows.

Economic message: Negative language contains incremental information about future fundamentals.



(a) Figure 1: Media coverage around earnings announcements.

Table I
Predicting Earnings Using Negative Words

This table shows estimates of the ability of negative words ($neg_{-30,-3}$) to predict quarterly earnings (SUE or $SAFE$) using ordinary least squares (OLS) regressions. We display the regression coefficients and summary statistics from six regressions below: two different dependent variables (SUE and $SAFE$) regressed on negative words computed based on four sets of news stories (*Dow Jones News Service (DJNS)*, *The Wall Street Journal (WSJ)*, "Before Forecasts," and "All Stories"). SUE is a firm's standardized unexpected quarterly earnings; and $SAFE$ is the standardized analysts' forecast error for the firm's quarterly earnings. The negative words variable ($neg_{-30,-3}$) is the standardized number of negative words in the news stories from 30 to 3 trading days prior to an earnings announcement divided by the total number of words in these news stories. The *DJNS* and *WSJ* regressions use only stories from these sources to compute $neg_{-30,-3}$. The two "Before Forecasts" regressions compute $neg_{-30,-3}$ using only stories that occur 1 trading day before the most recent consensus analyst forecast. All regressions include control variables for lagged firm earnings, firm size, book-to-market, trading volume, recent and distant past stock returns, and analysts' quarterly forecast revisions and dispersion (see text for details). Following Froot (1989), we compute clustered standard errors by calendar quarter. The robust t -statistics are in parentheses.

Stories Included	<i>SUE</i>				<i>SAFE</i>	
	<i>DJNS</i>	<i>WSJ</i>	Before Forecasts	All Stories	Before Forecasts	All Stories
$neg_{-30,-3}$	-0.0584 (-4.42)	-0.1083 (-5.28)	-0.0640 (-3.95)	-0.0637 (-4.69)	-0.0192 (-3.79)	-0.0197 (-4.44)
<i>Lag(Dependent Var)</i>	0.2089 (11.82)	0.2082 (11.83)	0.2042 (11.90)	0.2101 (11.98)	0.2399 (7.82)	0.2523 (8.74)
<i>Forecast Dispersion</i>	-0.9567 (-9.84)	-1.0299 (-9.59)	-0.9634 (-9.21)	-0.9373 (-10.20)	-0.2984 (-5.34)	-0.3076 (-6.34)
<i>Forecast Revisions</i>	20.2385 (8.89)	18.0394 (7.91)	20.4855 (8.51)	19.5198 (8.94)	0.5111 (0.68)	0.7580 (1.19)
<i>Log(Market Equity)</i>	-0.0071 (-0.40)	0.0003 (0.01)	-0.0043 (-0.24)	-0.0037 (-0.21)	0.0258 (4.79)	0.0289 (5.32)
<i>Log(Book / Market)</i>	0.0173 (0.62)	0.0182 (0.56)	0.0221 (0.77)	0.0204 (0.75)	-0.0162 (-1.97)	-0.0110 (-1.41)
<i>Log(Share Turnover)</i>	-0.1241 (-3.09)	-0.1348 (-2.90)	-0.1095 (-2.75)	-0.1261 (-3.20)	0.0274 (2.69)	0.0254 (2.61)
$FFA_{\alpha-252,-31}$	1.9784 (9.14)	1.9711 (9.90)	1.9770 (10.01)	2.0015 (9.50)	0.2199 (4.17)	0.2382 (4.36)
$FFCAR_{-30,-3}$	0.0119 (6.76)	0.0129 (6.33)	0.0117 (6.28)	0.0116 (6.64)	0.0062 (10.17)	0.0071 (11.38)
$FFCAR_{-2,-2}$	0.0104 (1.65)	0.0103 (1.37)	0.0177 (2.40)	0.0118 (1.91)	0.0053 (2.30)	0.0037 (1.89)
Observations	16755	11192	13722	17769	12907	16658
Clusters	80	79	78	80	78	79

(b) Table I: Predicting earnings using negative words.

Interpretation: The key comparison is that the negative-word coefficient remains significant after analyst and return controls. This means language is not redundant with quantitative information.

6.2 Figure 2: Firms' Fundamentals around Positive and Negative News Stories

Read: Positive-news firms have rising cumulative SUE; negative-news firms have falling cumulative SUE.

Economic message: The figure visually supports the claim that language is related to future cash-flow fundamentals.

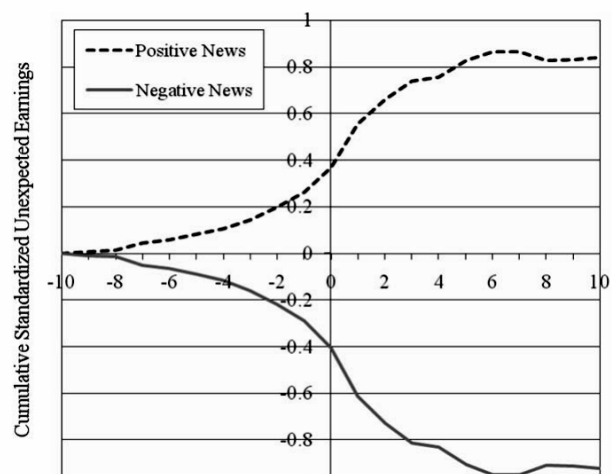


Figure 2: Firms' fundamentals around positive and negative news stories.

Interpretation: The post-event divergence is consistent with negative words summarizing information about future earnings rather than only past performance.

6.3 Table II: Predicting Returns Using Negative Words; Figure 3: Firms' Valuations around Positive and Negative News Stories

Read: Table II shows negative words predict lower next-day returns, especially for DJNS. Figure 3 shows quick day-0 reaction plus small continuation.

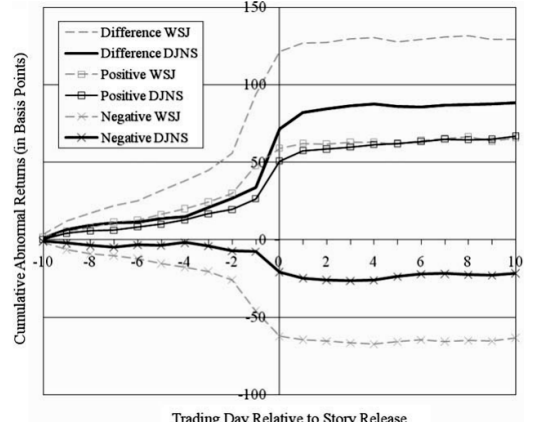
Economic message: Prices are attentive to language but slightly underreact at very short horizons.

Table II
Predicting Returns Using Negative Words

This table shows the relationship between standardized fractions of negative words (*neg*) in firm-specific news stories and firms' stock returns on the following day ($Return_{+1,+1}$ or $FFCAR_{+1,+1}$). The coefficients on $neg_{-30,-3}$ and summary statistics from six regressions are displayed below: two different dependent variables ($Return_{+1,+1}$ and $FFCAR_{+1,+1}$) regressed on negative words from each of three sets of news stories (*Dow Jones News Service [DJNS]*, *The Wall Street Journal [WSJ]*, and all stories). We exclude stories that occur after 3:30 pm (30 minutes prior to market closing). We assume that *WSJ* stories printed in the morning's *WSJ* are available to traders before the market close on the same day. The two dependent variables are the firm's raw close-to-close return ($Return_{+1,+1}$) and the firm's abnormal return ($FFCAR_{+1,+1}$). We use the Fama-French three-factor model with a $[-252,-31]$ trading day estimation period relative to the release of the news story as the benchmark for expected returns. The key independent variable is *neg*, the fraction of negative words in each news story standardized using the prior year's distribution. Each regression also includes control variables for the firm's most recent earnings announcement (*SUE*), market equity, book-to-market equity, trading volume, and close-to-close returns on the day of the news story, each of the previous two trading days, and the previous calendar year. Following Froot (1989), we compute clustered standard errors by trading day. The robust *t*-statistics are in parentheses.

Stories Included	$Return_{+1,+1}$			$FFCAR_{+1,+1}$		
	DJNS	WSJ	All	DJNS	WSJ	All
<i>neg</i>	-0.0277 (-3.67)	-0.0105 (-1.24)	-0.0221 (-3.72)	-0.0320 (-4.83)	-0.0102 (-1.37)	-0.0253 (-4.88)
$FFCAR_{0,0}$	0.0285 (5.28)	0.0229 (2.92)	0.0246 (5.43)	0.0259 (5.00)	0.0224 (2.94)	0.0226 (5.19)
$FFCAR_{-1,-1}$	-0.0272 (-3.63)	-0.0154 (-2.17)	-0.0222 (-4.21)	-0.0254 (-3.86)	-0.0106 (-1.68)	-0.0190 (-4.13)
$FFCAR_{-2,-2}$	-0.0215 (-3.16)	-0.0094 (-1.10)	-0.0179 (-3.39)	-0.0207 (-3.10)	-0.0104 (-1.22)	-0.0183 (-3.60)
$FFCAR_{-30,-3}$	-0.0005 (-0.30)	0.0016 (0.73)	-0.0002 (-0.13)	0.0004 (0.28)	0.0018 (0.85)	0.0005 (0.38)
$FFAlpha_{-252,-31}$	0.0559 (0.57)	0.1470 (1.29)	0.1046 (1.27)	0.1201 (1.36)	0.1686 (1.67)	0.1465 (2.02)
<i>Earnings (SUE)</i>	0.0160 (2.84)	0.0082 (1.33)	0.0125 (2.68)	0.0152 (3.46)	0.0055 (1.09)	0.0115 (3.25)
<i>Log(Market Equity)</i>	-0.0152 (-2.02)	-0.0159 (-1.99)	-0.0154 (-2.39)	-0.0120 (-2.19)	-0.0121 (-1.97)	-0.0109 (-2.51)
<i>Log(Book/Market)</i>	-0.0027 (-0.18)	0.0087 (0.60)	-0.0010 (-0.08)	-0.0246 (-2.12)	-0.0061 (-0.52)	-0.0201 (-2.22)
<i>Log(Share Turnover)</i>	-0.0324 (-1.66)	-0.0278 (-1.43)	-0.0300 (-1.76)	-0.0189 (-1.35)	-0.0167 (-1.16)	-0.0145 (-1.27)
Observations	141541	84019	208898	141541	84019	208898
Clusters (Days)	6260	6229	6272	6260	6229	6272
Adjusted R^2	0.0024	0.0014	0.0018	0.0026	0.0014	0.0019

(a) Table II: Predicting returns using negative words.



(b) Figure 3: Firms' valuations around positive and negative news stories.

Interpretation: The DJNS-WSJ contrast is useful because DJNS is more timely; the pattern points to short-run processing frictions.

6.4 Table III: Risk-Adjusted News-Based Trading Strategy Returns; Figure 4: Distribution of Daily Abnormal Returns

Read: The long-positive/short-negative DJNS strategy earns significant daily alphas before costs. Figure 4 shows positive abnormal returns in most years.

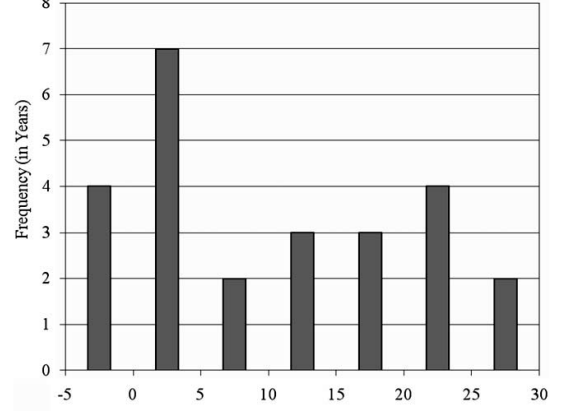
Economic message: The return continuation is systematic enough to survive risk adjustment before trading costs.

Table III
Risk-Adjusted News-Based Trading Strategy Returns

This table shows the daily risk-adjusted returns (*Alpha*) from a news-based trading strategy for three different time periods (1980 to 1994, 1995 to 2004, and 1980 to 2004). The first three regressions use the Fama-French (1993) three-factor model to adjust the trading strategy returns for the impact of contemporaneous market (*Market*), size (*SMB*), and book-to-market (*HML*) factors. The last three regressions use the Carhart (1997) four-factor model to account for incremental impact of the momentum factor (*UMD*). Table III reports the alpha and loadings from the time-series regression of the long-short news-based portfolio returns on each of the four factors. We assemble the portfolio for the trading strategy at the close of each trading day. We form two equal-weighted portfolios based on the content of each firm's *Dow Jones News Service* stories during the prior trading day. We label all news stories with a fraction of negative words in the previous year's top (bottom) quartile as negative (positive) stories. We include all firms with positive news stories in the long portfolio and all firms with negative news stories in the short portfolio. We hold both the long and short portfolios for one full trading day and rebalance at the end of the next trading day. We exclude the rare days in which there are no qualifying firms in either the long or the short portfolio. We compute all coefficient standard errors using the White (1980) heteroskedasticity-consistent covariance matrix. The robust *t*-statistics are in parentheses.

	1980–1994	1995–2004	1980–2004	1980–1994	1995–2004	1980–2004
<i>Alpha</i>	0.0919 (2.83)	0.1175 (3.93)	0.1031 (4.55)	0.0952 (2.81)	0.1131 (3.78)	0.1013 (4.38)
<i>Market</i>	−0.0994 (−0.93)	−0.1087 (−1.99)	−0.0983 (−1.86)	−0.0831 (−0.75)	−0.1001 (−1.87)	−0.0999 (−1.87)
<i>SMB</i>	−0.0767 (−0.35)	0.0475 (0.70)	−0.0081 (−0.08)	−0.0647 (−0.29)	0.0341 (0.49)	−0.0128 (−0.12)
<i>HML</i>	−0.1869 (−1.24)	−0.2590 (−2.81)	−0.2372 (−2.94)	−0.1819 (−1.20)	−0.2500 (−2.75)	−0.2365 (−2.93)
<i>UMD</i>				−0.0911 (−0.74)	0.0930 (2.01)	0.0444 (0.90)
Trading Days	3398	2497	5895	3398	2497	5895
Adjusted R^2	0.0003	0.0081	0.0026	0.0004	0.0106	0.0027

(a) Table III: Risk-adjusted news-based trading strategy returns.



(b) Figure 4: Distribution of daily abnormal returns.

Interpretation: This is a calendar-time diagnostic, not proof of easy arbitrage. It shows systematic delayed reaction.

6.5 Table IV: Sensitivity of News-Based Trading Returns to Trading Cost Assumptions

Read: At zero costs, abnormal annualized returns are 23.17%; at 8 bps they fall to 1.99%; at 9–10 bps they become negative.

Economic message: Trading costs can eliminate profits, making the anomaly economically modest.

Interpretation: Statistical predictability need not imply implementable abnormal returns.

6.6 Table V: Predicting Earnings Using Negative Words about Fundamentals

Read: The interaction $neg \times Fund$ is strongly negative for both SUE and SAFE.

Economic message: Negative words are most informative when the story is about fundamentals.

Interpretation: This is the central mechanism test. Generic negativity matters less than earnings-related negativity.

6.7 Table VI: Firms' Returns and Negative Words about Fundamentals

Read: Earnings-related negative words generate larger same-day reactions and larger next-day continuations.

Table IV
Sensitivity of News-Based Trading Returns
to Trading Cost Assumptions

This table shows estimates of the impact of transaction costs on the news-based trading strategy's profitability (see the text or Table III for strategy details). We recalculate the trading strategy returns for 11 alternative assumptions about a trader's round-trip transaction costs: 0, 1, 2, 3 ... or 10 basis points (bps) per round-trip trade. The abnormal and raw annualized cumulative news-based strategy returns for each assumption appear below. The risk-adjustment is based on the full-sample Fama-French three-factor loadings of the news-based portfolio shown in Table III.

Trading Costs (bps)	Abnormal Annualized Returns (%)	Raw Annualized Returns (%)
0	23.17	21.07
1	20.30	18.25
2	17.50	15.49
3	14.76	12.80
4	12.09	10.17
5	9.47	7.60
6	6.92	5.09
7	4.43	2.64
8	1.99	0.25
9	-0.39	-2.09
10	-2.71	-4.37

Table IV: Sensitivity of news-based trading returns to trading costs.

Table V
Predicting Earnings Using Negative Words about Fundamentals

This table reports the results from two OLS regressions with different dependent variables (*SUE* and *SAFE*) regressed on negative words ($neg_{-30,-3}$), fundamental words ($Fund_{-30,-3}$), and the interaction between these words ($neg_{-30,-3} * Fund_{-30,-3}$). *SUE* is a firm's standardized unexpected quarterly earnings; *SAFE* is the standardized analysts' forecast error for the firm's quarterly earnings. Both regressions include all news stories from both news sources (*Dow Jones News Service* and *The Wall Street Journal*) over the time period from 1984 through 2004. The measure of negative words ($neg_{-30,-3}$) is the standardized fraction of words that are negative in the news stories from 30 trading days prior up to 3 trading days prior to an earnings announcement. Fundamental words ($Fund_{-30,-3}$) is the fraction of words that are contained in news stories that mention the word stem "earn" from 30 trading days prior up to 3 trading days prior to an earnings announcement. All regressions include control variables for lagged firm earnings and numerous firm characteristics (see text for details). To allow for correlations among announced firm earnings within the same calendar quarter, we compute clustered standard errors (Froot (1989)). The robust *t*-statistics are in parentheses.

	<i>SUE</i>	<i>SAFE</i>
$neg_{-30,-3}$	-0.0167 (-1.19)	-0.0072 (-1.65)
$neg_{-30,-3} * Fund_{-30,-3}$	-0.3192 (-8.00)	-0.0824 (-5.48)
$Fund_{-30,-3}$	-0.4676 (-7.27)	-0.1033 (-5.59)
<i>Lag(Dependent Var)</i>	0.2080 (12.22)	0.2517 (8.69)
<i>Forecast Dispersion</i>	-0.9280 (-10.32)	-0.3049 (-6.35)
<i>Forecast Revisions</i>	19.1856 (9.06)	0.7068 (1.11)
<i>Log(Market Equity)</i>	-0.0062 (-0.36)	0.0285 (5.31)
<i>Log(Book / Market)</i>	0.0126 (0.48)	-0.0127 (-1.60)
<i>Log(Share Turnover)</i>	-0.1086 (-2.89)	0.0299 (3.08)
$FFAlpha_{-252,-31}$	1.9760 (9.59)	0.2317 (4.30)
$FFCAR_{-30,-3}$	0.0102 (5.74)	0.0067 (11.20)
$FFCAR_{-2,-2}$	0.0110 (1.81)	0.0036 (1.87)

Table V: Predicting earnings using negative words about fundamentals.

Economic message: Investors recognize fundamental stories but still underreact slightly.

Table VI
Firms' Returns and Negative Words about Fundamentals

This table shows the relationship between negative words in firm-specific news stories (*neg*) and firms' close-to-close abnormal stock returns on the same day ($FFCAR_{+0,+0}$) and the following day ($FFCAR_{+1,+1}$). The stories include all *Dow Jones News Service* articles from 1980 through 2004, but exclude stories that occur after 3:30 pm (30 minutes prior to market closing). The coefficients and summary statistics from two OLS regressions using two different dependent variables ($FFCAR_{+0,+0}$ and $FFCAR_{+1,+1}$) appear below. The key independent variable is negative words (*neg*), which is the fraction of negative words in each news story standardized using the prior year's distribution. The independent variable *Fund* is a dummy indicating whether a story mentions the word stem "earn"; and *neg * Fund* is the interaction between negative words (*neg*) and this dummy. All regressions include numerous control variables for lagged firm returns and other firm characteristics (see text for details). To allow for correlations among firms' returns within the same trading day, we compute clustered standard errors (Froot (1989)). The robust *t*-statistics are in parentheses.

	$FFCAR_{+0,+0}$	$FFCAR_{+1,+1}$
<i>neg</i>	-0.0857 (-11.16)	-0.0161 (-2.35)
<i>neg * Fund</i>	-0.3127 (-10.75)	-0.1036 (-4.52)
<i>Fund</i>	-0.3250 (-12.84)	-0.0342 (-1.96)
$FFCAR_{+0,+0}$		0.0255 (4.91)
$FFCAR_{-1,-1}$	0.0181 (2.23)	-0.0256 (-3.89)
$FFCAR_{-2,-2}$	-0.0220 (-2.72)	-0.0209 (-3.13)
$FFCAR_{-30,-3}$	0.0020 (1.17)	0.0004 (0.23)
$FFAlpha_{-252,-31}$	-0.1425 (-1.36)	0.1136 (1.29)
<i>Earnings (SUE)</i>	0.0234 (4.16)	0.0146 (3.33)
Observations	141633	141541
Clusters	6260	6260
Adjusted R^2	0.0045	0.0028

Table VI: Firms' returns and negative words about fundamentals.

Interpretation: The market processes language quickly, but fundamentals-related negative tone still predicts a small continuation.

7 Short summary

- Negative words in firm-specific news predict lower earnings and lower next-day returns.
- The earnings result survives analyst forecasts, past returns, and firm controls.
- The return result is strongest for DJNS and is short lived.
- A news-based long-short strategy has positive risk-adjusted returns before transaction costs.
- Transaction costs can erase the trading profits.
- Predictability is strongest for stories mentioning earnings, supporting a fundamentals interpretation.

8 Conclusion

8.1 Core conclusion

Simple dictionary-based language measures help predict firm fundamentals and short-run price movements.

8.2 Economic intuition

Qualitative news descriptions summarize hard-to-quantify information about firms. Investors incorporate most of this information quickly, but a small part is absorbed with a one-day delay.

8.3 Incremental contribution in retrospect

The paper became a foundational benchmark for textual finance because it showed that even simple, transparent word counts can forecast both cash-flow outcomes and price discovery.

8.4 Limitations

The dictionary does not understand context, negation, sarcasm, or firm-specific meanings; the design is predictive rather than randomized; and trading profits are sensitive to costs.

8.5 Takeaway

Words are data. Quantified qualitative information can improve the measurement of fundamentals and the study of market efficiency.